

A9 · Who Are You Really Talking To?

CS425 Computer Networks · instructor led · every command runs on Onyx (`ssh onyx`)

TURN THIS IN ON PAPER
before you leave the room
graded pass/fail

Names _____ Date _____

Every round: **predict** first, in ink, then **run** the command with the projector, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

ROUND 1 · DID ANYONE CHANGE THIS? `sha256sum` · `openssl dgst -hmac`

QUESTION	ANSWER
Predict: <code>\$10</code> becomes <code>\$90</code> . How many of the 64 hex digits change?	
Roughly how many actually changed?	

Q1. Trudy changes `$10` to `$90` and sends a fresh plain hash with it. Can Bob tell? Why does the same trick fail against the MAC?

ROUND 2 · WHO VOUCHES FOR WHOM? `openssl s_client` · `openssl x509`

QUESTION	ANSWER
Predict: who issued Boise State's certificate, and for how long is it good?	
Issuer, and the <code>notBefore</code> to <code>notAfter</code> span in days	
Whose name is in the subject? Where does <code>www.boisestate.edu</code> appear instead?	

DEPTH	CERTIFICATE FOR (s:)	SIGNED BY (i:)
0		
1		
2		

Q2. Why does Onyx believe this certificate belongs to Boise State? Who is Onyx really trusting at the end of the chain, and where did that trust come from?

ROUND 3 · FOUR WAYS TO FAIL `curl https://*.badssl.com/`

SITE	PREDICT: REFUSED?	WHAT <code>curl</code> SAID WAS WRONG	BROKEN LINK IN THE CHAIN
expired			
wrong.host			
self-signed			
untrusted-root			

Q3. With `-k` the connection is still encrypted. What exactly did `-k` give up, and which property from section 8.1 is that?

ROUND 4 · WHAT TLS DOES NOT HIDE `curl -sv · openssl s_client -trace`

Q4. You are on coffee shop Wi-Fi and every site you visit uses HTTPS. List everything the person running the Wi-Fi still learns about you. Aim for at least four things.

EXIT QUESTION · THE YES YOU TYPED IN A2 `ssh-keygen -lf · ssh-keygen -F`

QUESTION	ANSWER
Do the two fingerprints match? (first 8 characters after SHA256:)	

Exit. When you typed `yes` in A2, what were you trusting? How is that different from the way `curl` decided to trust `www.boisestate.edu`?

Graded pass/fail on something in every box. Hand this in before you leave.